

Migrating the Trading Desk backend from WSGI (Bottle) to ASGI (FastAPI)

Repository	trading-desk, branch hw-5.5-asgi-migration
Machine	Apple M3 (4 fast + 4 efficient cores), 16 GB, macOS 27.0, mains power
Docker	Docker Desktop 29.4, Linux VM with 8 CPUs and 8 GB
Image	python:3.14.7-slim: Debian 13, Python 3.14.7
Stack	Bottle 0.13.4, gunicorn 26.2.0, SQLAlchemy 2.0.52, psycopg 3.3.4, PostgreSQL 18.6, Docker Compose
Part 2 variant 3	FastAPI 0.141.1, uvicorn 0.53.0, a2wsgi 1.10.10

- **Exact Python version:** 3.14.0–3.14.4 use an incremental garbage collector; 3.14.5 returned to three generations. Memory and latency figures apply to 3.14.7.
- **Slim image:** 44 MB download instead of 404 MB (arm64). No compiler needed: every dependency installs from a prebuilt wheel. Services and benchmark share the same base image.

1. Summary

Decision: keep Bottle for request handling; serve the two live streams (prices and valuations) with FastAPI. Recommended as the next step.

- **One desk process per service keeps 100 traders' screens within budget on gunicorn, 200 with FastAPI serving the streams** (10 000 instruments, 5 000 open positions, live trading).
- **At 200 traders, valuations reach the screen in about 1 s instead of 2 s**, and market-data + pricing use 37 % less CPU (0.64 vs 1.02 cores).
- **Orders take 0.11–0.15 s in every variant**, also with 20 traders trading at once.
- **Part 1** (a books-service sample): the framework alone brings nothing; threads and open streams set the limit. Stage 4B replaced the `wsgiref` development server with gunicorn.

2. System

Six Bottle services, one PostgreSQL database, a React UI behind the Vite proxy. 43 endpoints: appendix A.

2.1 How the services run

Server	gunicorn, one gthread worker per service (stage 4B); before it, <code>wsgiref</code> , a development server
Threads	<code>SERVER_THREADS</code> , default 40: one thread per request or open stream; market-data, pricing and monitoring serve streams and use 256
Processes	One worker per service: services keep live state in memory
Database	SQLAlchemy ORM, synchronous sessions, psycopg 3; 15 connections per service ($6 \times 15 = 90$, under PostgreSQL's default 100)
Calls to other services	<code>urllib</code> : blocking, with timeouts
Errors	JSON <code>{"error": "..."} </code> for every status

Service (port)	Endpoints	Does
market-data (8001)	14	Polls market-data APIs, stores quotes and curves, streams updates
pricing (8002)	8	Revalues trades on every market update; valuations, book risk, previews
monitoring (8003)	5	Checks every service, collects and streams logs
books (8004)	6	Trading books
blotter (8006)	7	Trades with valuations, book summaries, audit history
trade-action (8008)	3	Validates and executes trade orders, one transaction per action

2.2 Load character

- **All six are I/O-bound** (from the code and 61 000 provider calls in the logs).
- **Request handlers:** short database queries. The only computation: pricing's `POST /price` and `POST /scenario`.
- **Long waits run in background threads** (external APIs, other services' streams, health polling). The framework does not change them.
- **External APIs inside a request:** market-data's `GET /symbols/search` and `POST /refresh`; they answer in 83–378 ms at the median, up to 1.8 s at p95.
- **Streams to the UI:** market-data, pricing, monitoring; every open UI holds two or three.
- **A migration can change only:** database waits in handlers, the UI streams, market-data's in-request API calls.

2.3 Dependencies under ASGI

No library needs WSGI; only our own code blocks an async migration.

Dependency	Under ASGI
Bottle 0.13	replaced by FastAPI
SQLAlchemy 2.0	async sessions in <code>sqlalchemy.ext.asyncio</code> ; needs greenlet, already installed
psycopg 3	async connections built in
structlog	request context lives in <code>contextvars</code> ; async code keeps it per task
Alembic	runs migrations in its own container, outside the services
desk-pricing	pure Python, no dependencies
Our code	blocking <code>urllib</code> calls; streams fed by background threads; handlers use Bottle's global <code>request</code> and <code>response</code>

2.4 Simplified before measuring

	Before	After
Python (services + libs)	11 964 lines, 139 files	10 206 lines, 129 files (–15 %)
Frontend (js / jsx)	13 376 lines	12 381 lines (–7 %)
Endpoints	50	43 (market-data –5, trade-action –3, pricing +1)
Migrations	9 files	1 baseline
market-data	5 010 lines	3 957 lines
blotter	848 lines	~590 lines

- **Streams:** three hand-written SSE copies → one `EventHub` and `follow_stream` in `desk_runtime/streams.py`; each frame serialized once.

- **market-data**: one owner of "is this symbol served" (`quote_board.py`); three feed copies → `SymbolQuoteFeed`; no full active-set query per quote.
- **pricing**: no `FOR UPDATE` transaction per tick; valuation status, `stale_at`, curve hints and curve moves in bp computed in the backend.
- **trade-action**: no queue, worker or second validation; one transaction per action; 201 / 200 (replay) / 422.
- **blotter**: no second in-memory copy of trades; 5 queries per page instead of 5 + N; totals and currency conversion in the backend.
- **monitoring**: one loop per target; state at once, audit after 3 failures.
- **UI**: no curve maths, currency totals, freshness rules or statuses; dead `sessionStorage` cache removed.
- **Infra**: `python:3.14.7-slim` pinned; one compose block and one `HEALTHCHECK`.

3. Method

Rules for both parts:

- **Criteria committed before the first run** of each part: [docs/decision_criteria.md](#) (Part 1: 174eb35; Part 2: c10e7dd; Part 2b: 1749779).
- **Machine**: one laptop (header); services, load and database in Docker on the same VM.
- **Pinned versions**: every package in `requirements.txt`; appendix B.
- **Production mode**: no reload, no debug, no access log.
- **Variants interleaved** within each run.
- **Each point = median of 3 runs; spread = max – min.**
- **A difference counts only if larger than the larger spread.**
- **More than 1 % errors or timeouts → the variant loses the point.**

3.1 Part 1: the sample

books-service code and a stub standing in for another service, behind Bottle and FastAPI; identical responses. Request → short database query → JSON, like nearly every handler; no background threads, nothing in memory.

Variant	Server	Handlers and clients
<code>bottle-sync</code>	gunicorn, sync worker	one request at a time
<code>bottle-threads</code>	gunicorn, 40 threads	<code>requests</code> , sync database driver
<code>bottle-threads-256</code> †	gunicorn, 256 threads	as <code>bottle-threads</code>
<code>bottle-wsgiref</code> †	wsgiref, the pre-4B server	as <code>bottle-threads</code>
<code>fastapi-async</code>	uvicorn	<code>async def</code> , <code>httpx</code> , async database driver
<code>fastapi-sync</code>	uvicorn	<code>def</code> in a 40-thread pool, <code>requests</code> , sync database driver

#	Request	Shows	In this system
S1	<code>GET /health</code>	framework and server overhead	every request
S2	<code>GET /io</code> : the stub waits 50 ms	waiting on another service	market-data's symbol search and refresh
S3	<code>GET /cpu</code> : 20 000 × SHA-256	computation	pricing's <code>POST /price</code>
S4	insert a book, read it by id	synchronous database work	most handlers
S5	<code>GET /health</code> at c = 10 beside 10, 50, 200 open streams	whether open streams block other requests	the UI streams

- **Load**: c = 1, 10, 50, 200; target c = 50. Latency budget p95 ≤ 100 ms (S3 none).

- **One point:** 10 s warm-up, 30 s measured; client timeout 10 s; load generator oha.
- **Setup:** one process per variant, server on its own CPU; S4 on a separate PostgreSQL, emptied before each point.
- **Metrics:** successful req/s; p50, p95, p99; errors and timeouts; server CPU and RSS.

Fixes made during the measurement:

Problem	Fix
htpx spent 20 % of the CPU on a missing package (<code>sniffio</code>)	package installed
htpx with an unlimited connection pool collapsed at $c = 200$	default pool
Warm-up requests still running during the measurement	warm-up waits for them
Load generator ran out of ports: <code>bottle-sync</code> closes every connection	port reuse; first full run discarded
Database pool kept reopening connections, FastAPI 2–4× more often	15 connections kept open; S4 measured again

3.2 Part 2: the desk core

The business core of the desk: quotes in, valuations out, trades through.

#	Variant	Serving
1	<code>wsgiref</code>	the development server before stage 4B
2	<code>gunicorn</code>	<code>gthread</code> , one worker, $\text{threads} = 40 + K + 10$ for market-data and pricing (stage 4B)
3	FastAPI streams	FastAPI serves <code>/stream</code> (market-data) and <code>/valuation-stream</code> (pricing); their other routes are the same Bottle app behind <code>a2wsgi</code> (40 workers); the other four services as in variant 2

- **Same application code** in all variants; only the HTTP layer differs.
- **Desk:** 10 000 watched equities; 1 000 of them held, refreshed every 15 s, the rest every 60 s (about 217 quotes per second); 5 000 open positions (3 500 spot, 500 bonds, 500 swaps, 500 options).
- **Provider stub:** constant 200 ms, seeded random-walk prices; 401 req/s at p99 203 ms alone.
- **Watching clients:** $K = 10, 50, 100, 200$; each holds the price stream and the valuation stream, about 390 frames per second. $K = 200$ means about 78 000 frames per second.
- **Trading:** 0.2 orders opened per second (720 per hour), each closed about 60 s later. Part 2b: $T = 5$ and 20 traders at once, one order about every 20 s each.
- **Price previews:** `POST /price`, a European option, 0.2 per second.
- **One run:** fresh database from a template → all 10 000 quotes and 5 000 valuations live → 90 s warm-up → clients connect at $K / 10$ per second → 30 s without a dropped stream → 150 s measured.
- **Each service on its own CPU of the VM:** market-data 7, pricing 6, trade-action and stub 5, blotter 4, the rest 2–3, load client 0–1.
- **A run counts only if** all clients stay connected, the load client uses < 0.8 cores and other programs on the laptop use ≤ 3 cores during the window. No run crossed a limit.

Budget (p95)	
Tick delivery: provider response → tick at the client	≤ 1 s
Valuation freshness: provider response → valuation at the client	≤ 1 s
Price preview (<code>POST /price</code>)	≤ 300 ms
Position on screen: order confirmed → first valuation of that position	≤ 3 s
Order execution (<code>POST /trade-actions</code>)	≤ 500 ms
Errors: timeouts, 5xx, dropped streams	≤ 1 %

4. Part 1: the sample (books-service)

Successful req/s / p95 ms at c = 50 (S5: 50 open streams). p99 in brackets where the tail is the result.

	bottle- sync	bottle- threads	bottle- threads-256 †	bottle- wsgiref †	fastapi- async	fastapi- sync
S1 /health	11 479 / 4.7	14 116 / 5.7	14 013 / 4.9	7 016 / 0.9	33 064 / 2.5	17 447 / 3.2
S2 /io	17.5 / 2 886	711 / 105	918 / 59.2	656 / 56.2 (p99 1 094)	693 / 90.5	719 / 101
S3 /cpu	201 / 256	201 / 555	202 / 496	207 / 1 103 (p99 2 526)	216 / 380	212 / 460
S4 /db	1 117 / 48.5	1 343 / 120	1 346 / 72.6	1 249 / 19.0 (p99 1 050)	1 094 / 73.2	1 343 / 57.6
S5 /health	–	0 / all timeouts	13 367 / 0.8	7 058 / 0.9	29 074 / 0.4	–
S2, c = 200	5.3, 71.5 % errors	705 / 302	2 966 / 84.5	1 700 / 55.2 (p99 1 308)	360 / 606	700 / 314
S3, c = 200	203 / 1 004	202 / 1 445	194 / 2 109	207 / 1 894, 4.4 % errors	210 / 613, 3.8 % errors	210 / 1 206

† One run, 24 September. The others: median of 3 runs; spread typically 1–7 % of the median, S4 up to 25 %. Every range: `benchmark/results/summary.md`, `benchmark/results/threads-check/summary.md`. –: not measured.

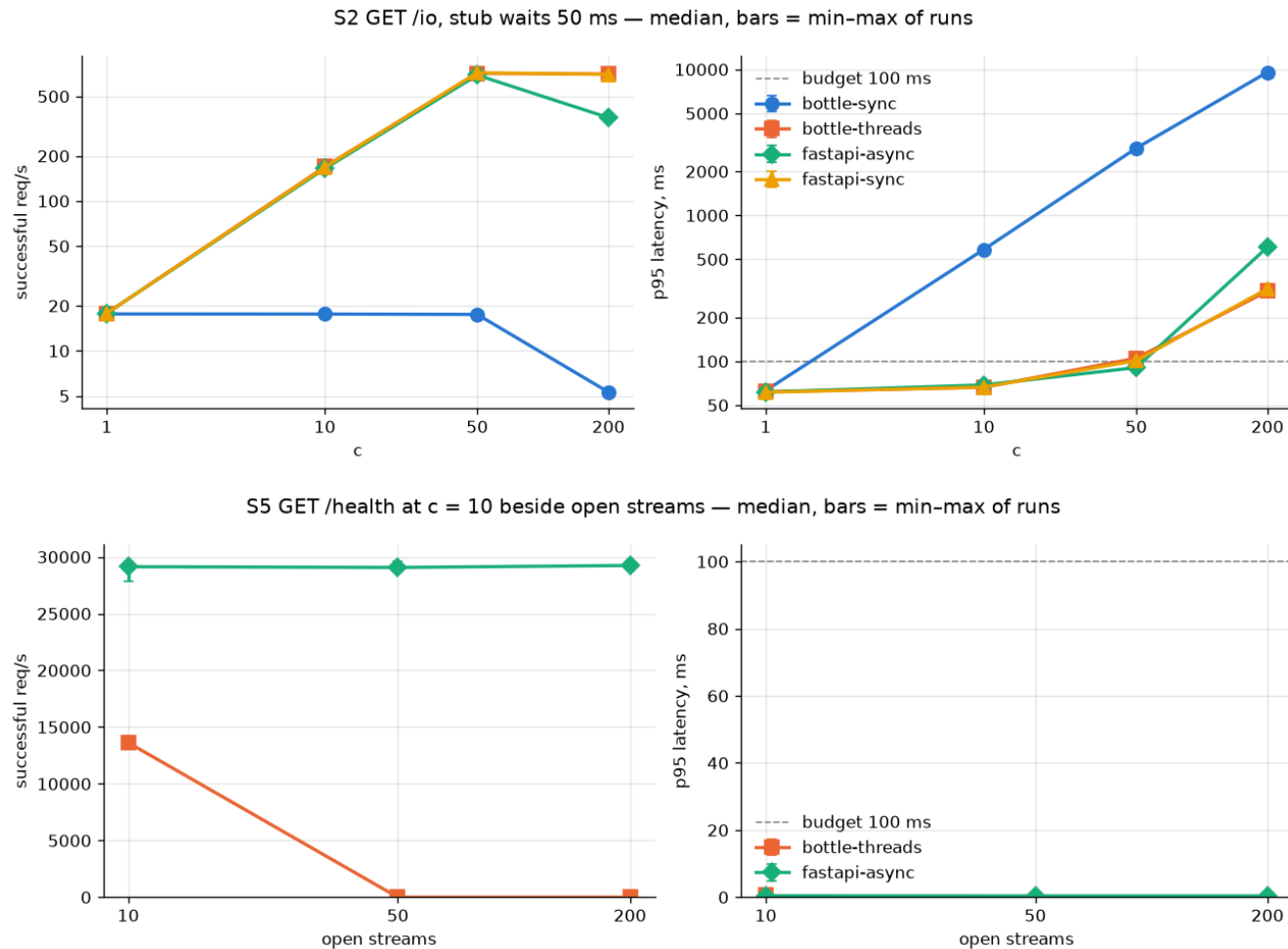
Observations

- **Open streams exhaust a fixed thread pool:** beside 50 streams `bottle-threads` (40) answers nothing; with 256 threads 13 367 req/s at 0.8 ms; `fastapi-async` 29 074 req/s at 0.4 ms.
- **Waiting is limited by threads, not by the framework:** at 40 threads `Bottle` and `FastAPI` `def` both stop near 710 req/s (711, 719); with 256 threads `Bottle` reaches 918 req/s at p95 59.2 ms, and 2 966 req/s at c = 200, where `fastapi-async` gives 360.
- `wsgiref` **has a 1 s tail:** p99 1 094 ms in S2 and 1 050 ms in S4 at c = 50, from c = 10 (S2 p99 1 079 ms); its listen queue of 5 drops connections, which retry after 1 s. p95 hides it.
- **Computation is the same everywhere:** S3 201–216 req/s on one core; only the order of service changes p95 (256 to 555 ms).
- **A short database request gains nothing from async:** S4 `FastAPI` `def` = `Bottle` threads (1 343 req/s each); `fastapi-async` 1 094.
- **FastAPI async has the lowest framework overhead:** S1 33 064 req/s vs 11 479–17 447; every variant answers in 0.1 ms at c = 1.
- **Memory:** `Bottle` 77–87 MB, `FastAPI` 110–117 MB.

Result of Part 1 (criteria 174eb35): NO-GO for a full migration

Gate	Result
1. No regression in S3, S4	FAIL: S3 p95 <code>FastAPI</code> 380 ms, <code>Bottle</code> 256 ms
2. Real gain in S2 or S5	PASS: S5, <code>Bottle</code> answers nothing beside 50 streams
3. Cost	not scored
4. Need: <code>Bottle</code> over budget, <code>FastAPI</code> within	PASS at 40 threads; FAIL at 256 threads (S2 p95 59.2 ms)

- **What Part 1 could not say:** whether ASGI's one win, open streams, matters at the desk's real volume (Part 2).



All charts, S1–S5: [benchmark/results/charts/](#) (3 runs, min–max bars) and [benchmark/results/threads-check/charts/](#) (†).

Limitations

- One laptop; macOS decides whether the server's CPU is a fast or a slow core.
- Each client waits for its answer before the next request: overload latency is understated.
- The stub waits a fixed 50 ms; the real external APIs take 83–378 ms.
- The sample has no background threads and no in-memory state.

5. Part 2: the desk core

p95 in ms, median of 3 runs (spread in brackets). *X* = over budget. Every table and run: [benchmark/results/desk_day/summary.md](#).

K watching clients	10	50	100	200
Valuation freshness ($\leq 1\,000$)				
wsgiref	189	303	1 029 <i>X</i>	2 419 <i>X</i>
unicorn	177	368	719	1 960 <i>X</i>
FastAPI streams	176	300	913	959

K watching clients	10	50	100	200
Tick delivery ($\leq 1\ 000$)				
wsgiref	47.4	60.9	81.0	103
gunicorn	47.5	61.2	74.8	110
FastAPI streams	48.2	57.3	69.8	84.1
Price preview (≤ 300)				
wsgiref	152	175	237	467 ✗
gunicorn	152	205	223	394 ✗
FastAPI streams	183	170	237	286
Position on screen ($\leq 3\ 000$)				
wsgiref	2 426	2 414	2 482	3 077 ✗
gunicorn	2 408	2 419	2 622	2 573
FastAPI streams	2 451	2 387	2 465	2 484
Order execution (≤ 500)				
wsgiref	119	114	121	129
gunicorn	125	122	123	124
FastAPI streams	125	131	137	122
Largest K within budget	wsgiref 50	gunicorn 100	FastAPI streams 200	

- **Errors:** 0 % everywhere; no dropped stream in 36 runs.

Observations

- **Up to about 100 clients the server makes no measurable difference:** at K = 100 FastAPI's freshness spans 426–1 437 ms over three runs and overlaps the others.
- **At 200 clients FastAPI keeps valuations fresh; threads do not.** Every FastAPI run (743 / 959 / 1 107 ms) beats every gunicorn run (1 807 / 1 960 / 2 727 ms).
- **The same shows on the server:** pricing's lag behind market data at K = 200 is 0.7–1.3 s with FastAPI, 1.6–2.9 s with gunicorn.
- **Why:** on gunicorn every open stream is a thread woken for each frame: 200 threads in market-data at about 170 frames a second, 200 in pricing at about 220. They compete for the interpreter with pricing's single revaluation thread. FastAPI serves all streams from one event loop.
- **Tick delivery and previews improve less** (84 vs 110 ms, 286 vs 394 ms): –24 % and –27 %.
- **Orders do not depend on the variant:** 114–137 ms; FastAPI does not change trade-action.
- **The margin is thin:** FastAPI's K = 200 freshness median is 959 ms against a 1 000 ms budget; one run was 1 107 ms.

Part 2b: many traders trading at once (K = 50; [benchmark/results/desk_day_trading/summary.md](#))

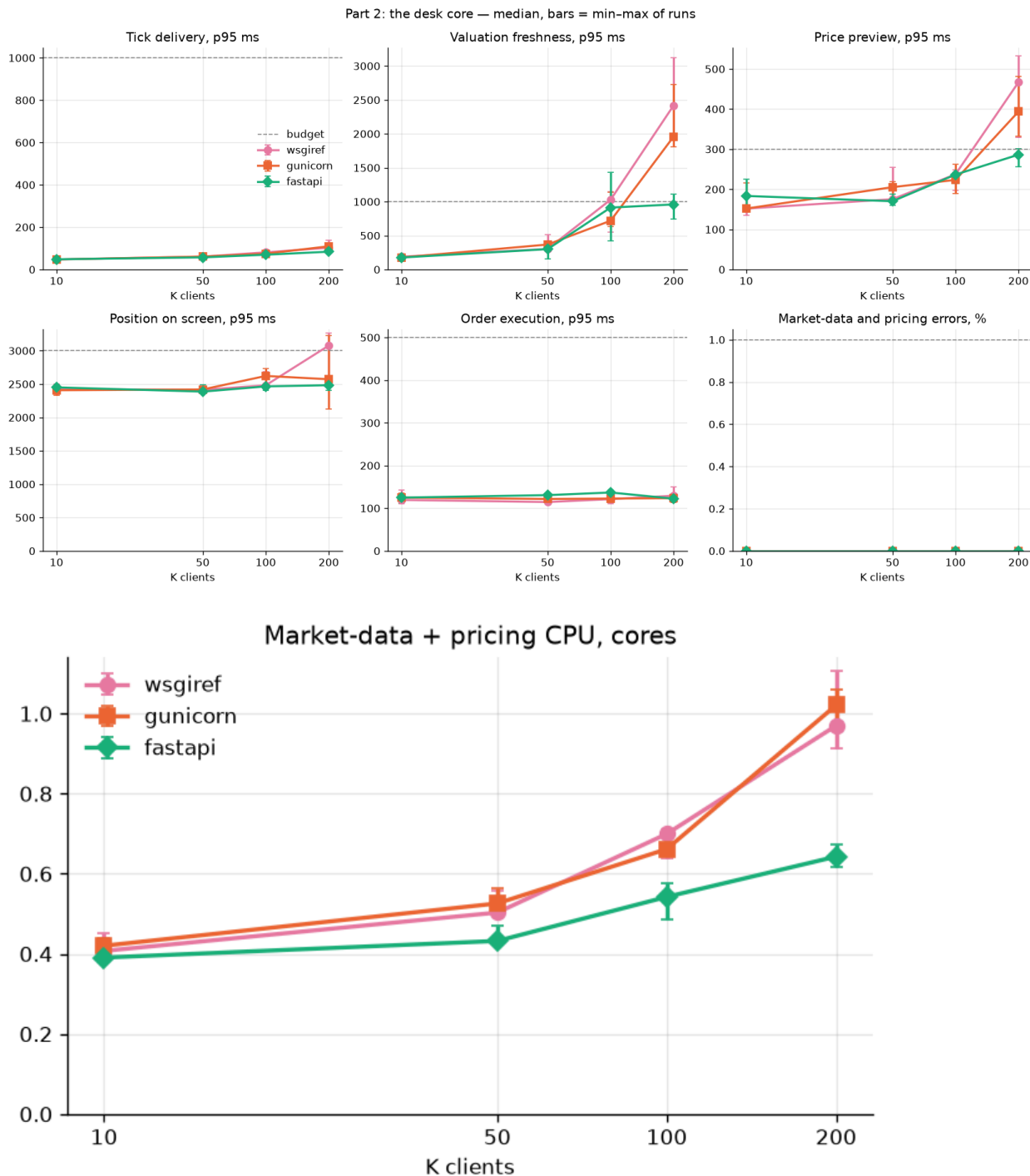
T traders	5 (0.25 orders/s)			20 (1 order/s)		
	wsgiref	gunicorn	FastAPI	wsgiref	gunicorn	FastAPI
Order execution	124	119	130	150	138	137
Valuation freshness	334	440	201	413	376	258
Position on screen	2 481	2 527	2 430	2 421	2 409	2 385

- **Every variant within every budget;** 0 errors; about 150 orders opened per run at T = 20.
- **Orders take the same time in every variant:** differences are within the spread (up to 53 ms).

- The Part 2 decision stands (criterion 1749779).

What limits the desk besides the server (measured on gunicorn while setting up Part 2)

- **Logging in costs 12.8 MB per trader:** /snapshot 7.4 MB and 0.9 s of market-data CPU, /valuations 5.4 MB.
- **A price preview checks the underlying against the whole watchlist:** 120–170 ms of pricing CPU; one preview per second already pushes freshness past 1 s with one client.
- **Every new trade makes pricing rebuild its index of 5 000 positions** (every 2 s): at 2 orders per second freshness reaches 1 s with one client.
- **The Trades & PnL screen returns 8.2 MB in 290–480 ms:** polled every 5 s, it fills the blotter's core at about 15 traders (calculated).



Part 2b charts: benchmark/results/desk_day_trading/charts/.

Limitations

- The whole desk on one laptop; other programs used a median of 1.3 cores during the runs.
- p95 of orders and previews rests on about 30 samples per run.
- Clients hit the services directly, not through the Vite proxy; they do not poll screens.
- One process per service in every variant; many processes behind a shared message bus were not measured.

6. CPU cost

Cloud cost follows cores.

Per request (Part 1, $c = 50$): CPU ms per request, which is also the cores one service needs for 1 000 requests per second.

	Bottle, 40 threads	FastAPI def	FastAPI async
S1 /health : framework only	0.07	0.06	0.03
S2 /io : waits on a service	0.45	0.43	0.73 (httpx)
S3 /cpu : computation	4.97	4.71	4.62
S4 /db : database	0.74	0.74	0.91 (async driver)

Per trader (Part 2): market-data + pricing cores (per 100 watching clients).

K	10	50	100	200
wsgiref	0.41	0.50	0.70	0.97 (0.48)
gunicorn	0.42	0.53	0.66	1.02 (0.51)
FastAPI streams	0.39	0.43	0.54	0.64 (0.32)

- **Async libraries cost more per request:** +63 % CPU when a request waits on a service, +23 % on the database. Moving request handlers to async would raise the bill.
- **Async streams cost less per trader:** at $K = 200$ market-data drops from 0.62 to 0.36 cores and pricing from 0.40 to 0.28; together 37 % less.
- **In business terms:** 200 traders' screens need about one core for market-data and pricing on gunicorn and about two thirds of a core with FastAPI streams.

7. Decision (ADR)

ADR-002 · Serve the market-data and pricing streams with FastAPI · 29 September 2026

Decision: GO for the streams, NO-GO for the rest. FastAPI serves market-data's `/stream` and pricing's `/valuation-stream`; every other route stays Bottle; the other four services stay on gunicorn.

Why

- **The criterion passed on two counts:** FastAPI streams stay within the market-data and pricing budgets at 200 clients, gunicorn at 100 (a); valuation freshness at $K = 200$ is 51 % lower, beyond the spread (b).
- **It costs less CPU, not more:** −37 % for market-data + pricing at $K = 200$.
- **It holds under trading:** 5 and 20 traders trading at once change nothing (Part 2b).

- **Request handlers stay synchronous:** Part 1 showed no gain for short database requests and higher CPU for async libraries.

Context: six Bottle services; ADR-001 (Part 1, 24 September) chose gunicorn threads over a full migration. Does ASGI help the desk's core at real volume?

Criteria: docs/decision_criteria.md, Part 2 (c10e7dd) and Part 2b (1749779), committed before the runs.

Criterion	Result
(a) One K level higher within the market-data and pricing budgets	yes: FastAPI 200, gunicorn 100
(b) p95 ≥ 30 % lower at K = 200, beyond the spread	yes, freshness: 959 vs 1 960 ms; tick -24 % and preview -27 % do not count
CPU per client $\leq +10$ %	yes: -37 %
Part 2b: the decision holds with many traders	yes: all variants within budget at T = 5 and 20

- **Uncertainty:** (a) rests on a median of 959 ms against 1 000 ms; (b) and the CPU saving are clear of the spread.

Options

Option	For	Against
1. Keep gunicorn everywhere	one runtime; enough up to about 100 clients	thread-per-stream; freshness over 1 s at 200 clients
2. FastAPI for the streams (chosen)	200 clients within budget; -37 % CPU; 107 lines in the proof of concept	two runtimes; a bridge (a2wsgi) for the other routes
3. Full async migration	one framework	no gain for database requests; $+23$ – 63 % CPU per request; 43 endpoints to port

8. Risks

What could go wrong with option 2 in production. P = probability, I = impact: L / M / H.

#	Risk	P	I	Mitigation	Warning signal
1	A blocking call on the event loop freezes every stream. Only the stream hub runs on the loop; any synchronous code added to a native route stops all clients	M	H	native routes only for streams; review rule; a test holding a stream while a slow route runs	tick delivery p95 > 1 s while CPU < 50 %
2	The bridge becomes the new thread limit. Other routes run in a2wsgi's 40 workers; previews cost 120–170 ms CPU each	M	M	size the workers; make preview validation cheap	preview p95 > 300 ms
3	The gain is smaller than expected. Freshness is limited by pricing's rebuild of 5 000 positions, not only by the server	M	M	fix the rebuild before scaling; measure again	freshness p95 > 1 s with pricing CPU < 50 %
4	Slow clients handled differently. Overflow closes an async queue instead of EventHub's sentinel	L	M	the same overflow test for both hubs	overflow warnings without reconnects

#	Risk	P	I	Mitigation	Warning signal
5	Two runtimes to operate. gunicorn and uvicorn differ in shutdown, logging and settings	H	L	one launcher in <code>desk_runtime</code> ; same 5 s graceful stop and log format	<code>docker stop</code> > 10 s; log lines missing in the Logs view
6	New dependencies. FastAPI, Starlette, uvicorn, <code>a2wsgi</code> (a small project)	M	M	pinned versions; the bridge can later be replaced by native routes	advisories against a pinned version
7	Error format drifts. FastAPI answers <code>{"detail": ...}</code> by default	L	L	only two native routes; a check that every error has an <code>error</code> key	error responses without <code>error</code> key
8	Scope creeps into async database code. Part 1: no gain, +23 % CPU	M	M	scope agreed: streams only	an async database driver added

- **Top 3:** 1, 2, 3.

9. Alternative plan (stage 4B) and next step (stage 4A)

Stage 4B, after Part 1's NO-GO: gunicorn instead of `wsgiref` — implemented (3790d5d)

- gunicorn gthread, one worker per service; threads from `SERVER_THREADS`; startup work and background threads in the worker; 5 s graceful stop; streams send a comment every 5 s so a closed client frees its thread.

	Before (<code>wsgiref</code>)	After (gunicorn)
Sample, S2 at c = 50 (Part 1)	656 req/s, p99 1 094 ms	918 req/s, p99 61.7 ms (256 threads)
Desk: clients within budget (Part 2)	50	100
Desk: valuation freshness at K = 100	1 029 ms	719 ms
<code>/health</code> beside 200 streams	1 121 req/s, p95 12.7 ms	6 393 req/s, p95 2.4 ms
<code>docker stop</code>	10.23 s, killed	0.41–0.83 s; 5.31–5.40 s with open streams; clean exit
Closed stream clients	threads kept until the next event (395 threads)	thread freed within 5 s

- **Cost:** 10–13 MB more memory per container; open streams are capped by the thread count (256 held 255 streams, then waited).

Stage 4A, after Part 2's GO: recommended next step

- Move variant 3 from `benchmark/desk_day/fastapi_streams.py` into `desk_runtime`: a launcher for market-data and pricing; FastAPI, uvicorn and `a2wsgi` pinned; README; the full UI check.
- Expected effect (Part 2): 200 instead of 100 clients within budget; –37 % market-data + pricing CPU at 200 clients.

Revisit when

- the desk needs more than about 200 screens on one process per service;
- services must run as several processes (then streams move to a shared message bus);
- pricing's rebuild and the preview validation are fixed, and freshness is limited only by the server.

10. Conclusions

Answers to the assignment's questions

1. **Problem and whether it exists:** thread-per-stream serving. Not at today's load (one UI, two or three streams); it appears from about 200 watching clients on one desk.
2. **What the scenarios measure:** Part 1, one request type at a time on a sample; Part 2, the desk core at 10 000 instruments.
3. **Uncertainty:** median of 3 runs. Up to 100 clients the variants overlap. At 200 the freshness ranges do not overlap; (a) rests on a median 41 ms under budget.
4. **What decided:** Part 2 criterion (a) and (b) on valuation freshness, with 37 % less CPU. It would change under the conditions below.
5. **Top 3 risks:** a blocking call on the event loop, the bridge's thread limit, a gain smaller than expected (section 8).
6. **Cost against the estimate:** the proof of concept is 107 lines; the forecast gave FastAPI 50–200 clients and GO a 40 % chance; the result is at the top of that range.
7. **What instead, and its effect:** after Part 1, unicorn: no 1 s tail, clean stops, 100 instead of 50 clients on the desk (section 9).
8. **Lessons:**
 - **Benchmarks:** measure the application's own costs first (previews, logins and screens limited the desk before the server did); keep the load client and other programs out of the result (checked each run); warm up; alternate variants.
 - **Async:** it pays where many connections wait (open streams), not in short database requests; its libraries cost more CPU per request.

What would change the result

- **Many processes:** in-memory state keeps each service to one process; with a shared cache and a message bus (e.g. Redis pub-sub) streams could be spread over processes, and the server model would matter less.
- **A cheaper pricing engine:** rebuilding 5 000 positions and validating previews against 10 000 instruments limit freshness in every variant.

Appendix A. Endpoints

All responses are JSON. Errors are `{"error": "..."}` with the HTTP status — also for unknown routes (404), wrong methods (405) and unhandled exceptions (500). SSE endpoints return `text/event-stream`. `/health` answers `{service, status}`; pricing and blotter add counters.

market-data-service (14)

Method	Path	Params	Codes	Body
GET	<code>/stream</code>	—	200	SSE: quote and curve updates
GET	<code>/snapshot</code>	—	200	<code>{stream_id, event_id, spots, curves}</code>
GET	<code>/curves, /curves/<provider></code>	<code>raw</code>	200, 404	list of curves
GET	<code>/curves/<provider>/<curve>/<as_of></code>	<code>raw</code>	200, 400, 404	one curve
GET	<code>/quotes/<provider>/<symbol>/history</code>	<code>limit 1–200, raw</code>	200, 400, 404	list of snapshots

Method	Path	Params	Codes	Body
GET	/watchlist	—	200	list of items
POST	/watchlist	JSON body	201, 400, 409, 422	item
DELETE	/watchlist/<symbol>	provider	200, 404, 409	removal result
GET	/fx/rates	to	200, 400	{to, rates}
GET	/symbols/search	q (≥ 2 chars)	200, 400	{query, results, provider_errors}
GET	/providers	—	200	provider status
POST	/refresh	symbol, provider	200, 404, 422, 429, 502, 503	tick, or {refreshed, skipped} without symbol
GET	/health	—	200	{service, status}

pricing-service (8)

Method	Path	Params	Codes	Body
GET	/curves/<curve>/at	maturity_years, index_tenor 3M/6M or payments_per_year	200, 400, 404	zero rate, discount factor, par rate
GET	/valuations	—	200	list of valuations
GET	/valuations/<trade_id>	—	200, 404	one valuation
GET	/book-risk	—	200	list of book-risk records
POST	/price	JSON body	200, 400, 404, 409, 503	priced instrument
POST	/scenario	JSON body	200, 400, 404	scenario result
GET	/valuation-stream	—	200	SSE: valuation and book-risk updates
GET	/health	—	200	{service, status, ...}

monitoring-service (5)

Method	Path	Params	Codes	Body
GET	/status	—	200	health of every target
GET	/audits	limit, since, severity, service, event_type, correlation_id, entity_id	200, 400	list of audit rows
GET	/logs	level, service, q, limit (1–10 000)	200, 400	{lines, meta}
GET	/logs/stream	—	200	SSE: run, then log lines
GET	/health	—	200	{service, status}

books-service (6)

Method	Path	Params	Codes	Body
GET	/books	—	200	list of books
GET	/books/<book_id>	—	200, 404	one book
POST	/books	name (≤ 60), expected_asset_class, description (≤ 200)	201, 400, 409	created book
PUT	/books/<book_id>	partial body	200, 400, 404, 409	updated book
DELETE	/books/<book_id>	—	200, 404, 409	deactivated book
GET	/health	—	200	{service, status}

trade-action-service (3)

Method	Path	Params	Codes	Body
GET	/instruments/term-schemas	—	200	{instruments, schemas, curves}
POST	/trade-actions	JSON: OPEN_TRADE, CLOSE_TRADE, REASSIGN_TRADES	201, 200 (replay, close, reassign), 400, 422	result
GET	/health	—	200	{service, status}

blotter-service (7)

Method	Path	Params	Codes	Body
GET	/trades, /trades/overview	limit 1–500, offset, book_id, asset_class, status, symbol	200, 400	list of trades / {trades, books}
GET	/trades/<trade_id>	—	200, 404	trade with valuations and audit logs
GET	/trades/<trade_id>/valuations, .../audit-logs	—	200, 404	list
GET	/books/summary	currency	200, 400	book summaries
GET	/health	—	200	{service, status, ...}

Appendix B. Commands and versions

Part 1

	Command
bottle-sync	gunicorn -w 1 sample_wsgi.app:app
bottle-threads	gunicorn -w 1 -k gthread --threads 40 sample_wsgi.app:app

	Command
fastapi-async	uvicorn sample_asgi.app:app --no-access-log
fastapi-sync	uvicorn sample_asgi.app:sync_app --no-access-log
Stub	uvicorn downstream_stub.app:app --no-access-log (one process)
Load, per point	oha -z 10s -w -c <c> -t 10s <url> (warm-up), then oha -z 30s -c <c> -t 10s --output-format json <url>

```
benchmark/run_benchmark.sh          # full grid, about 2 h 45 min
SCENARIO5=s4 benchmark/run_benchmark.sh  # one scenario again
docker compose -f benchmark/infra/compose.yml down  # removes the benchmark database
```

Part 2

```
benchmark/run_desk_day.sh          # K = 10 50 100 200, 3
variants, 3 runs, about 3.5 h
RESULTS=results/desk_day_trading KS=50 TRADERS="5 20" benchmark/run_desk_day.sh # Part 2b, about 1.5 h
docker compose -f benchmark/infra/desk.compose.yml run --rm client python -m desk_day.analyze_desk
docker compose -f benchmark/infra/desk.compose.yml down -v          # removes the desk database
```

- **Versions:** Python 3.14.7, Bottle 0.13.4, gunicorn 26.2.0, FastAPI 0.141.1, uvicorn 0.53.0 (uvloop 0.22.1, httptools 0.8.0), a2wsgi 1.10.10, httpx 0.28.1, requests 2.34.2, SQLAlchemy 2.0.52, psycopg 3.3.4, PostgreSQL 18.6, oha 1.16.0. All pins: benchmark/infra/requirements.txt; services: requirements.txt.
- **Machine:** Apple M3 (4 fast + 4 efficient cores), 16 GB, macOS 27.0, mains power; Docker Desktop 29.4, VM with 8 CPUs and 8 GB. Part 1: server on CPU 7, stub 6, PostgreSQL 4–5, load 0–3.
- **Part 1 settings** (benchmark/infra/compose.yml): 15 database connections kept open, log level WARNING, load generator may reuse closed ports.
- **Part 2 settings** (benchmark/infra/desk.compose.yml): 15 database connections per service, log level WARNING, provider limits lifted, 64 quote requests in flight.